

Bhasura Suseen Gunawardhana

<https://bhasurasuseen.github.io/>

Education

Texas A&M University — College Station, TX, USA

Doctor of Philosophy in Visual Computing & Interactive Media —GPA 4.0/4.0 (Expected Graduation May 2028)

Aug 2024 - Present

University of Houston — TX, USA

Master of Science in Computer Science — GPA 3.934/4.0

Aug 2022 - Aug 2024

University of Colombo School of Computing (UCSC) — Colombo 07, Sri Lanka

Bachelor of Science in Computer Science — GPA 3.39/4.0

Feb 2015 - Feb 2019

Royal College Colombo — Colombo 07, Sri Lanka

GCE A/L in Physical Science (Combined Mathematics, Physics, Chemistry)

May 2010 - Dec 2013

GCE O/L (Mathematics, Science and Technology, History, English, Art, Literature, Geography, Health Science, Religion)

June 2004 - Dec 2009

Research Interests

Interactive Computer Graphics, Human-Computer Interaction(HCI), Extended Reality (XR), Visualization, Virtual Production, Animation Synthesis, Motion Capture, Volume Capture, Digital Twins, Gaming, Game-Based Learning, Deep Learning, Simulation, Agent-Based Modeling, Computer Vision, and Robotic Simulation.

Technical Skills

Software Development & Programming:

C++, Python, C#, R, Java, JavaScript, Unreal Engine, Unity3D, OpenGL, GLSL, HLSL, VTK, OpenCV, PyTorch, Git, SVN, Linux

Creative Media & Design:

Houdini, Autodesk Maya, Blender, Autodesk 3DS Max, ZBrush, Motion Builder, Adobe Photoshop, Adobe Illustrator, Adobe XD, Davinci Resolve, Adobe Premiere Pro, Adobe After Effects.

Publications

Gunawardhana, B.S., Zhang, Y., Sun, Q. and Deng, Z., 2024, October. [Toward User-Aware Interactive Virtual Agents: Generative Multi-Modal Agent Behaviors in VR](#). In 2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR) (pp. 1068-1077). IEEE.

Research Experience

Research Assistant - Virtual Production Institute College of Performance, Visualization & Fine Arts Texas A&M University Aug 2024 - Present

- VVAI Lab: Developing a Mixed Reality Digital Twin Framework for Training, Accessibility, and Pipeline Optimization in Virtual Production.
- Assisted Summer Industry Course(SIC) 2025 by troubleshooting technical problems related to Unreal Engine Pipeline. Developed TechViz Tools. Conducted mini workshops on Virtual Scouting, Virtual Camera, Motion and Facial Capture.
- Assisted in the process of building and configuring the initial virtual production stages for VPI, including ICVFX workflows with a large LED volume, cinema cameras, motion tracking systems, and real-time rendering software.

VR Agent Conversation Animation Synthesis University of Houston — TX, USA

Aug 2022 - Aug 2024

- Captured multimodal conversation data that includes full body VICON motion capture, eye tracking, speech audio, and video.
- Developed an Unreal Engine and LLM based VR conversation agent system.
- Designed and trained body and eye motion synthesis Generative AI(CVAE-Conditional Variational Auto Encoder) model.

Final Year Undergraduate Research - Individual Project

Feb 2018 - Feb 2019

Thesis Topic: ["Simulation of climbing plants with twining behavior"](#)

- Developed an experimental setup to study the behavior of twining plants, creating a dataset of growth time-lapses and plant measurements.
- Proposed a novel collision-based twining plant simulation model, enhancing the biomechanical aspects of existing models.
- Conducted real-time simulations, evaluating both behavioral realism and computational efficiency.

Virtual Production Experience

"The Experiment", "Plunder" Summer Industry Course Virtual Production Projects

Summer 2025

Unreal Pipeline Technician / TA

Tools & Technologies: Unreal engine / ICVFX: Sony Verona LED Panels, Stype Camera Tracking, Sony VENICE/BURANO.

"Sixth Side" Test Shoot for a Virtual Production Shortfilm in Pre-Production

Spring 2025

Responsible for Writing, Directing, and Virtual Production Supervising

Tools & Technologies: Unreal engine / ICVFX: Sony Verona LED Panels, Vicon Camera Tracking, Sony BURANO / Davinci Resolve

"The Elves and the Shoemaker" Virtual Production Short

Fall 2024

Virtual Production Technician

Tools & Technologies: Unreal engine / ICVFX: Sony Verona LED Panels, Vicon Camera Tracking, Sony BURANO

Teaching Experience

Unreal Engine Technician/TA —Summer Industry Course(SIC) 2025 in Virtual Production— Texas A&M University , TX, USA Summer 2025

Instructional Assistant (IA) — Interactive Computer Graphics (COSC 4370/6372) — University of Houston, TX, USA Spring and Summer 2024

Instructional Assistant (IA) — Interactive Computer Graphics (COSC 4370) — University of Houston, TX, USA Fall 2023

Instructional Assistant (IA) — Software Design (COSC 4353/6353) — University of Houston, TX, USA Spring and Summer 2022

Instructional Assistant (IA) —Computer Science and Programming (COSC 1336) — University of Houston, TX, USA Fall 2022

Visiting Lecturer — Game Designing (MI 5015)— Academy of Design, Colombo, Sri Lanka Feb 2020 - July 2020

(affiliated with the Northumbria University, Newcastle, UK)

Work Experience

Senior Software Engineer

June 2021 - July 2022

Software Engineer

Feb 2019 - May 2021

SimcCentric Technologies, Orion City IT Park, Colombo 09, Sri Lanka.

SAF-TAC Project - Epic Mega Grant Recipient

May 2019 - July 2022

VR/Desktop multiplayer application designed specifically for military simulation and training with essential simulation functions such as Exercise Control, Scenario Authoring, and After-Action Review.

Tools & Technologies: C++ / Unreal Engine 4 / Unreal Blueprints / Blender

- Contributed to the development of map and content modding infrastructure.
- Contributed to iterative development of hybrid (animation and IK driven) 3-point tracked full-body VR avatar system.
- Developed 3D interaction systems and 3D user interfaces for VR users.
- Developed 3D/2D scenario authoring/replay tools and [user interfaces](#).
- Developed camera, navigation, and switching systems for vehicles and characters.
- Developed time of the day and weather control features.
- Created dynamic paths (aircraft and projectile) and shape drawing plugins.
- Initiated profiling, optimization processes, and multiplayer replication best practices to improve performance and concurrent player count.
- Integrated in-house haptic suit and ProTubeVR haptic weapon.
- Carried out logic/content optimizations and configurations to support multiple VR devices (Oculus Rift/Rifts, Oculus Quest, Samsung Odyssey, HTC Vive, Varjo XR).

SAF-Fires Project

Feb 2019 - April 2019

Standards-based, Image generator agnostic, and scalable Call-For-Fire and Close Air Support training simulation suitable for deployment in Joint Fires and accreditable Joint Terminal Attack Controller (JTAC) training simulators.

Tools & Technologies: C/ C++

- Integrated NVIDIA PhysX physics engine.
- Developed rotary-wing flight dynamic model capable of handling manual controls from standard joysticks.

Research & Development Intern

Aug 2017 - Jan 2018

Synopsys

Tools & Technologies: C/C++/python/Verilog/VHDL/shell script/Linux

- Participated in training sessions and exercises on C, C++, register-transfer level (RTL) languages (Verilog, VHDL), [electronic design automation \(EDA\) technologies](#), and ASIC/SoC pipelines.
- Implemented new LINT rules and RTL test cases for an electronic design and verification tool.
- Analysed and fixed memory issues and contributed to developing regression test infrastructure.
- Contributed to R&D work to create a "Machine learning framework to predict which LINT rules to run on an electronic design"

Awards

Selected for the "Spiralation" Startup Support Program — Bamboo Game — ICT Agency of Sri Lanka (ICTA)	2017
1st Runner-up — Measure Wire — Microsoft Imagine Cup Sri Lanka, Innovation Category	2016
Finalist — Game of Thoughts — Microsoft Imagine Cup Sri Lanka, Games Category	2016
2nd Runner-up — LOL Game — Microsoft Imagine Cup Sri Lanka, Games Category	2015
Best Direction, Best Script — Theater Play - " Right is Wrong " — Royal College Colombo Inter House Sinhala Drama Competition	2011

3D Modeling & Animation

" FrogSized " 3D animation short: Responsible for all aspects	Fall 2024
Tools & Technologies: Unreal engine / Maya / Houdini / Substance Painter / Davinci Resolve	
3D animation advertisement for a digital banking solution	May 2022
Tools & Technologies: Unreal engine / Blender / Adobe After Effects	

Technical Projects

Interactive VR Volume Rendering to explore medical data	Fall 2023
Tools & Technologies: Unreal Engine 5, C++, HLSL	
VR game " All Elephants Go to Heaven " developed for Epic Mega Jam 2020	Dec 2020
Tools & Technologies: Unreal Engine 4, C++	
Desktop multiplayer game " Under The Ice " developed for Unreal Spring Jam 2020	July 2020
Tools & Technologies: Unreal Engine 4, C++	
Hospital channelling system for Southern Lanka Hospitals	Jan 2018 – Aug 2018
Tools & Technologies: Node.js / Feathers / MongoDB / Angular 5 / Angular Material / Ngx Charts / AWS	
Smart institute management solution " SIMS "	Jan 2018 – Aug 2018
Tools & Technologies: Node.js / Feathers / MongoDB / Angular 5 / Angular Material / Ngx Charts / AWS	
Desktop multiplayer game " Bambo Game " developed with Spiralation Startup program	2017 - 2018
Tools & Technologies: Unreal Engine 4, C++	
Hospital office management system for Panadura base Hospital	2016
Tools & Technologies : HTML / CSS / JavaScript / D3.js / PHP / MySQL / Azure	
IOT solution " Measure Wire " mobile app and a device where we can take our everyday measurements	Feb 2016 – May 2016
Tools & Technologies: Arduino / Android / Java	
Two-player game based on cellular automata named " Game of Thoughts "	Feb 2016 – May 2016
Tools & Technologies : Unity Game Engine / C# / 3ds Max	
Mobile game named " LOL Game " e where players can make characters laugh by tapping	Feb 2015 – May 2015
Tools & Technologies : Unity Game Engine / C# / Adobe Illustrator	

Sessions & Workshops

Visiting session on Unreal Engine 4 for UCSC 3rd year game development module	May 2020
Vocational Training Institute, Colombo 06, Sri Lanka - Unreal engine 4 sessions for game development	Jan 2020
Informatics Institute of Technology, Colombo 05, Sri Lanka - Unreal Engine 4 sessions for game development	Dec 2019
Mentor at "Game Rush" game hackathon organized by UCSC IEEE student branch	Dec 2019
Session on UE4 game development at "Game Rush " organized by UCSC IEEE student branch	Nov 2019

Leadership

Senior Prefect - Royal College, Colombo 07

Sept 2012 - Dec 2013

Chairman - Royal College Colombo Astronomical Society

Jan 2011 - Dec 2011

Assistant Chairman - Royal College Colombo Sinhala Drama Society

Jan 2010 - Dec 2010

Hobbies

Traveling, Reading, Watching Movies, Story & Script Writing, Observational Astronomy, Archery, Gaming.